MAMI Human Milk Microbiome: Exploratory Statistical Analysis

Stage 1 — alpha diversity, beta diversity and study-level comparisons

1,128 breast-milk samples were matched one-to-one between the MAMI abundance matrix and accession metadata. The original matrix has 580 feature columns; 346 were non-zero at least once. Three obvious host/organelle markers (Mammalia, Chloroplast and Mitochondria) were excluded, leaving 344 features. Rows were renormalized to sum to one after filtering.

Methods

Alpha diversity: observed feature richness, Shannon entropy, Simpson 1−D and Pielou evenness. Beta diversity: Bray–Curtis dissimilarity and classical PCoA. Study-group alpha diversity was tested with Kruskal–Wallis tests. Study-group beta diversity was tested with one-factor PERMANOVA using 999 unrestricted permutations.

Main results

Mean richness = 21.78 (median 17; range 1–126); mean Shannon = 1.449; mean Simpson 1−D = 0.619.

Bray–Curtis PERMANOVA by study: pseudo-F = 90.43; R^2 = 0.326; permutation p = 0.001.

Alpha metric	Kruskal–Wallis H	p-value
Observed_richness	353.67	2.514e-73
Shannon	194.62	2.642e-39
Simpson_1_minus_D	258.68	5.710e-53

Study-level summary

Study	Country	n	Mean richness	Mean Shannon
AsnicarF_2017	Italy	5	4.8	0.650
Grant-BeurmannS_2022	Nigeria	566	15.0	1.578
KhineWWT_2020	Indonesia	45	47.8	1.646
LundgrenSN_2018	USA	154	37.3	1.631
RamaniS_2018	India	159	23.8	1.267
WilliamsJE_2019	USA	148	15.6	0.922
WuY_2021	China	51	41.0	1.465

Interpretation and limitations

The observed profiles differ across study groups. This is a study-associated signal, not a clean country effect: country is confounded with study, and protocols, populations, sample handling, sequencing and bioinformatic processing may differ. PERMANOVA can also respond to differences in within-group dispersion; a PERMDISP test and sensitivity analyses are needed.

The supplied values appear relative-abundance-like rather than raw read counts. Renormalization does not recover sequencing depth or absolute microbial load. Feature labels are operational taxonomic features; only three obvious host/organelle markers were excluded. A full taxonomy and contaminant review is still required. Maternal age and several clinical covariates are missing or extremely sparse, so adjusted clinical or causal claims are not supported.

Next: verify taxonomy and contaminants; run PERMDISP and prevalence-threshold sensitivity analyses; use compositional differential-abundance methods with study-aware models; then define a leakage-resistant ML target and study-aware validation.

Exploratory computational analysis only; not a clinical finding or diagnostic tool.